Software Requirements Specification

Operating System File Explorer

Version 1.0 | Based on IEEE 830 / ISO/IEC/IEEE 29148

Status: Draft for Review

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Revision History

Version	Date	Author	Description
0.1	Oct 2026	Author	Initial draft
1.0	Oct 2026	Author	Baseline for review

1. Introduction

1.1 Purpose

This document specifies the software requirements for the **OS File Explorer**, a graphical application that lets users browse, search, organize and manage files and folders on local, removable and network storage. It is intended for developers, testers, project managers and stakeholders.

1.2 Scope

The File Explorer provides a visual interface to the operating system's file system. It supports navigation, file operations (copy, move, delete, rename, create), search, previews, properties, tabs, and integration with the OS shell (context menus, clipboard, drag and drop, trash). It does not include file content editing, cloud-sync engines or backup software.

1.3 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
GUI	Graphical User Interface
Path	Textual location of a file or folder in the file system
Metadata	Attributes such as name, size, type, dates and permissions
ACL	Access Control List defining permissions on a file object
Trash	Recoverable holding area for deleted items (Recycle Bin)
Shell	The OS layer providing context menus, clipboard and associations

1.4 References

- IEEE Std 830-1998, Recommended Practice for SRS
- ISO/IEC/IEEE 29148:2018, Requirements Engineering
- WCAG 2.1 Accessibility Guidelines
- POSIX / NTFS / APFS file system documentation

1.5 Document Overview

Section 2 describes the product context, Section 3 lists functional requirements, Sections 4 to 6 cover interfaces, quality attributes and data, and Sections 7 to 9 provide use cases, constraints and acceptance criteria.

2. Overall Description

2.1 Product Perspective

The File Explorer is a core OS component that sits between the user and the file system layer. It communicates with the kernel via file system APIs and with other applications through the shell, clipboard and drag-and-drop services.

Layer	Responsibility
Presentation (GUI)	Windows, tabs, panes, views, dialogs, context menus
Application Logic	Navigation, operations queue, search, selection, history
Service Layer	Indexing, thumbnails, previews, trash, permissions
File System API	Kernel calls: open, read, write, rename, delete, stat, watch
Storage	Local disks, removable media, network shares

2.2 Product Functions (Summary)

- Browse folders in a hierarchical tree and content pane
- Create, rename, copy, cut, paste, move and delete files and folders
- Search by name, type, date, size and content
- Multiple views: list, details, icons, tiles, columns
- Preview files and view properties
- Tabs, split view, favorites and recent items
- Compress and extract archives
- Manage permissions and mounted drives

2.3 User Classes and Characteristics

User Class	Description	Expertise
General User	Everyday browsing and organizing of personal files	Low
Power User	Bulk operations, shortcuts, advanced search, path editing	High
Administrator	Manages permissions, system folders and network shares	High
Accessibility User	Uses screen readers, keyboard-only or high contrast	Varies

2.4 Operating Environment

- Desktop OS (Windows, macOS, Linux) on x86-64 or ARM64
- Minimum 2 GB RAM, 50 MB disk for application and cache
- Display resolution 1024x768 or higher with scaling support

2.5 Design and Implementation Constraints

- Must use native OS file system APIs and honor OS permissions
- Must follow OS UI guidelines and localization framework
- Destructive operations must not bypass OS security policies

3. System Features (Functional Requirements)

Priority: **H** = High (must), **M** = Medium (should), **L** = Low (could).

3.1 Navigation

Users shall be able to move through the file system quickly and predictably.

ID	Requirement	Pri
FR-NAV-01	The system shall display a navigation pane with a folder tree, favorites, drives and network locations.	H
FR-NAV-02	The system shall display the contents of the selected folder in the main pane.	H
FR-NAV-03	The system shall provide back, forward and up-one-level navigation with history.	H
FR-NAV-04	The system shall provide an editable address bar with breadcrumb navigation and path autocomplete.	H
FR-NAV-05	The system shall allow pinning folders to favorites and show a recent items list.	M
FR-NAV-06	The system shall support tabs and split (dual-pane) view.	M

3.2 File and Folder Operations

Users shall be able to manage file system objects.

ID	Requirement	Pri
FR-OPS-01	The system shall create new files and folders in the current location.	H
FR-OPS-02	The system shall rename an item inline and reject invalid or duplicate names with a clear message.	H
FR-OPS-03	The system shall support copy, cut and paste via menu, shortcut and drag and drop.	H
FR-OPS-04	The system shall move deleted items to Trash by default and support permanent delete with confirmation.	H
FR-OPS-05	The system shall restore items from Trash to their original location.	H
FR-OPS-06	The system shall show progress, speed and time remaining for long operations, with pause and cancel.	H
FR-OPS-07	The system shall handle name conflicts by offering replace, skip or keep both.	H
FR-OPS-08	The system shall support undo and redo for rename, move, copy and delete (to Trash).	M
FR-OPS-09	The system shall support batch rename using patterns.	L
FR-OPS-10	The system shall compress items to archives and extract common archive formats (ZIP, TAR, GZ, 7Z).	M

3.3 Views and Display

The display shall be customizable.

ID	Requirement	Pri
FR-VIEW-01	The system shall offer list, details, icons, tiles and column views.	H

ID	Requirement	Pri
FR-VIEW-02	The system shall sort by name, size, type, date modified and date created, ascending or descending.	H
FR-VIEW-03	The system shall group and filter items by attribute.	M
FR-VIEW-04	The system shall allow showing or hiding hidden files and file extensions.	H
FR-VIEW-05	The system shall display thumbnails for images, videos and documents.	M
FR-VIEW-06	The system shall provide a preview pane for supported file types.	M
FR-VIEW-07	The system shall remember view settings per folder.	L

3.4 Search

Users shall locate items efficiently.

ID	Requirement	Pri
FR-SRC-01	The system shall search by file name within the current folder and subfolders.	H
FR-SRC-02	The system shall support filters: type, size range, date range and attributes.	H
FR-SRC-03	The system shall support wildcards and optionally regular expressions.	M
FR-SRC-04	The system shall search file contents for indexed text formats.	M
FR-SRC-05	The system shall display results progressively and allow cancelling a search.	H
FR-SRC-06	The system shall allow saving frequent searches.	L

3.5 Properties, Permissions and Metadata

Users shall inspect and modify item attributes where permitted.

ID	Requirement	Pri
FR-PRP-01	The system shall display name, type, location, size, size on disk and timestamps.	H
FR-PRP-02	The system shall calculate folder size and item counts asynchronously.	H
FR-PRP-03	The system shall allow viewing and editing permissions and ownership subject to user rights.	M
FR-PRP-04	The system shall allow setting read-only and hidden attributes.	H
FR-PRP-05	The system shall allow choosing the default application for a file type.	M

3.6 Devices and Network

The system shall handle storage devices and shares.

ID	Requirement	Pri
FR-DEV-01	The system shall list internal drives, removable media and network shares with capacity and free space.	H
FR-DEV-02	The system shall allow safe ejection of removable media.	H
FR-DEV-03	The system shall allow mapping and connecting to network shares with credentials.	M
FR-DEV-04	The system shall automatically detect device connection and removal and refresh the view.	H

3.7 Shell Integration and Interaction

ID	Requirement	Pri
FR-SHL-01	The system shall provide a context menu for files, folders and empty space.	H
FR-SHL-02	The system shall open files with the associated default application.	H
FR-SHL-03	The system shall support standard keyboard shortcuts (copy, paste, delete, rename, select all, find).	H
FR-SHL-04	The system shall support multi-selection with mouse, Shift, Ctrl/Cmd and rubber-band selection.	H
FR-SHL-05	The system shall support drag and drop to and from other applications.	H
FR-SHL-06	The system shall automatically refresh when the file system changes externally.	H

4. External Interface Requirements

4.1 User Interface

- Main window: toolbar, address bar, search box, navigation pane, content pane, optional preview and details panes, status bar
- Status bar shall show item count, selected count and total selected size
- Dialogs for confirmation, conflicts, errors and properties shall be consistent and keyboard accessible
- Light, dark and high-contrast themes shall be supported
- All text shall be localizable with right-to-left language support

4.2 Hardware Interfaces

- Keyboard, mouse, touchpad and touch screen input
- Storage devices via OS drivers (HDD, SSD, USB, optical, SD)

4.3 Software Interfaces

Interface	Purpose
File System API	Read directory entries, metadata, perform file operations
File System Watcher	Receive change notifications for live refresh
Clipboard Service	Transfer file references for copy and paste
Shell Association Registry	Resolve default applications and icons
Search Index Service	Fast name and content queries
Security Subsystem	Authentication, ACL checks, elevation prompts

4.4 Communication Interfaces

- Network file protocols such as SMB, NFS and FTP/SFTP where supported by the OS
- Credentials shall be passed through the OS credential store

5. Non-Functional Requirements

ID	Category	Requirement
NFR-PERF-01	Performance	A folder with 10,000 items shall begin displaying within 1 second and be fully listed within 3 seconds.
NFR-PERF-02	Performance	The UI shall remain responsive (no freeze over 100 ms) during copy, search and size calculation.
NFR-PERF-03	Performance	Name search shall return first results within 500 ms using the index.
NFR-PERF-04	Resource Use	Idle memory shall not exceed 150 MB; CPU usage while idle shall be under 1%.
NFR-REL-01	Reliability	Interrupted operations shall leave no partial or corrupted files; copies shall be verified on completion.
NFR-REL-02	Reliability	The application shall recover from crashes and restore the previous session.
NFR-REL-03	Availability	Failure of one tab or operation shall not terminate other tabs or operations.
NFR-SEC-01	Security	All operations shall enforce OS permissions; elevation shall be explicit and logged.
NFR-SEC-02	Security	The system shall protect against path traversal and warn on executable or system file changes.
NFR-SEC-03	Security	Stored credentials shall be encrypted via the OS credential vault.
NFR-USE-01	Usability	Common tasks shall be completed by a new user in no more than 3 steps.
NFR-USE-02	Accessibility	The interface shall conform to WCAG 2.1 AA, with full keyboard operation and screen reader support.
NFR-USE-03	Usability	Error messages shall state the cause and a suggested action.
NFR-MNT-01	Maintainability	Code shall be modular with at least 80% unit test coverage on core logic.
NFR-POR-01	Portability	Core logic shall be separated from platform-specific code to ease porting.
NFR-SCA-01	Scalability	The system shall handle directories of up to 1,000,000 entries using virtualized lists.
NFR-COM-01	Compatibility	Support for NTFS, FAT32, exFAT, ext4, APFS and network file systems per platform.
NFR-LOC-01	Localization	Dates, numbers and sort orders shall follow the user's locale; Unicode file names shall be fully supported.

6. Data Requirements

6.1 Logical Data Entities

Entity	Key Attributes
FileItem	name, path, extension, type, size, created, modified, accessed, attributes, owner, permissions
Folder	name, path, itemCount, totalSize, parent
Drive	label, mountPoint, fileSystem, capacity, freeSpace, removable
Favorite	name, path, order
HistoryEntry	path, timestamp
OperationJob	id, type, sources, destination, status, progress, errors
TrashRecord	originalPath, deletedDate, size, trashId
UserSettings	viewMode, sort, showHidden, theme, columns, shortcuts

6.2 Data Retention

- Settings, favorites and session state persist per user profile
- Recent items and history are capped (default 100 entries) and clearable
- Search index is stored locally and rebuildable on demand

6.3 Privacy

- No file content or names shall be transmitted off the device without explicit user consent
- Indexing shall exclude user-designated folders

7. Use Cases

Use Case	UC-01: Copy Files to Another Folder
Actor	User
Precondition	A source folder is open and the user has read access.
Main Flow	1. User selects files. 2. User chooses Copy. 3. User navigates to destination. 4. User chooses Paste. 5. System copies files and shows progress. 6. System displays the new items.
Alternate / Exceptions	Name conflict: system prompts replace/skip/keep both. Insufficient space: system halts and reports. Access denied: system skips item and logs error.

Use Case	UC-02: Delete and Restore a File
Actor	User
Precondition	The file exists and the user has delete permission.
Main Flow	1. User selects file and presses Delete. 2. System moves it to Trash. 3. User opens Trash, selects item, chooses Restore. 4. System returns file to original path.
Alternate / Exceptions	Original folder missing: system recreates path or asks for a new location. Permanent delete: system requests confirmation.

Use Case	UC-03: Search for a File
Actor	User
Precondition	A search scope folder is open.
Main Flow	1. User enters text in search box. 2. System lists matching items progressively. 3. User applies filters. 4. User opens a result or its containing folder.
Alternate / Exceptions	No results: system displays a helpful empty state. User cancels: search stops and partial results remain.

Use Case	UC-04: Change File Permissions
Actor	Administrator
Precondition	User has rights to modify permissions.
Main Flow	1. Open Properties. 2. Select Security tab. 3. Edit permissions. 4. Apply. 5. System validates and saves.
Alternate / Exceptions	Elevation required: system prompts for credentials. Invalid change: system rejects with explanation.

Use Case	UC-05: Safely Eject USB Drive
Actor	User
Precondition	A removable drive is mounted.
Main Flow	1. User right-clicks the drive and selects Eject. 2. System flushes pending writes. 3. System unmounts and confirms safe removal.
Alternate / Exceptions	Drive in use: system lists blocking processes and offers retry.

8. Constraints, Assumptions and Dependencies

8.1 Assumptions

- The underlying OS provides stable file system and notification APIs
- Users have a valid account with appropriate permissions
- Indexing service is available for content search

8.2 Dependencies

- OS kernel and file system drivers

- Graphics and font rendering libraries
- Third-party codecs and libraries for previews and archives

8.3 Constraints

- Operation speed is bounded by device and network throughput
- Features depend on file system capabilities (e.g., permissions on FAT32 are limited)

9. Acceptance Criteria

- All High priority requirements are implemented and pass test cases
- Performance targets in Section 5 are met on reference hardware
- No critical or high severity open defects
- Accessibility audit passes WCAG 2.1 AA
- Security review confirms no privilege escalation or data loss paths

Requirements Traceability (Excerpt)

Requirement	Use Case	Test Case
FR-OPS-03, FR-OPS-06, FR-OPS-07	UC-01	TC-COPY-01..05
FR-OPS-04, FR-OPS-05	UC-02	TC-DEL-01..04
FR-SRC-01, 02, 05	UC-03	TC-SRC-01..06
FR-PRP-03	UC-04	TC-PRM-01..03
FR-DEV-02, 04	UC-05	TC-DEV-01..03

10. Appendices

Appendix A: Default Keyboard Shortcuts

Action	Windows/Linux	macOS
Copy / Cut / Paste	Ctrl+C / X / V	Cmd+C / X / V
Rename	F2	Enter
Delete	Delete	Cmd+Backspace
Permanent delete	Shift+Delete	Option+Cmd+Delete
New folder	Ctrl+Shift+N	Cmd+Shift+N
Search	Ctrl+F	Cmd+F
Address bar	Ctrl+L / Alt+D	Cmd+Shift+G
Back / Forward	Alt+Left / Right	Cmd+[/]
New tab / Close tab	Ctrl+T / Ctrl+W	Cmd+T / Cmd+W
Properties	Alt+Enter	Cmd+I
Undo	Ctrl+Z	Cmd+Z

Appendix B: Error Handling Summary

Condition	System Response
Access denied	Show message, offer elevation if applicable, continue with remaining items
Disk full	Stop operation, report required vs available space
Path too long / invalid characters	Reject with explanation of limits
File in use	Offer retry, skip or show locking process
Network disconnect	Pause, retry automatically, then report failure